```
(venv) PS D:\Learning\future_aws_agent_engineer\multi_agent_customer_service_system> python src/agent_orchestrator.py test
Running local agent test...

------------------------------------------------------------

Session: 4eb11a00 | Customer: CUST-001
Query: I want to return my wireless headphones from order ORD-27176
I'll help you process your return request. Let me start by initializing the session and then gathering the necessary information about your order.
Tool #1: initialize_session

Tool #2: route_to_inventory_agent

  ------------------------------------------------------------
  [ STEP 01 ]  INVENTORY AGENT
               Role  : Fetch order details and customer tier from DynamoDB
               Model : us.anthropic.claude-sonnet-4-5-20250929-v1:0
  ------------------------------------------------------------

Tool #1: check_order_status

Tool #2: get_customer_tier
```json
{
  "order_details": {
    "order_id": "ORD-27176",
    "customer_id": "CUST-001",
    "product_name": "Wireless Headphones Pro",
    "product_category": "Electronics",
    "quantity": "1",
    "price": "149.99",
    "order_date": "2026-10-06",
    "status": "delivered",
    "estimated_delivery": "2026-10-11",
    "tracking_number": "TRK100000100"
  },
  "customer_details": {
    "customer_id": "CUST-001",
    "name": "Alice Johnson",
    "tier": "Premium",
    "email": "alice@example.com",
    "account_created": "2021-03-15",
    "name": "Alice Johnson",
    "tier": "Premium",    "email"
    "email": "alice@example.com",
    "account_created": "2021-03-15",
    "total_orders": 47,
    "preferred_contact": "email"
  }
}
```
               Status : [COMPLETE] Elapsed: 7.4s (LLM may still stream final text below)
               State : DynamoDB[inventory_agent] version 0 -> 1
Great! I've retrieved your order information. Now let me route this to the Refund Agent to process your return request.
Tool #3: route_to_refund_agent

  [STEP 02] REFUND AGENT
               Role : Evaluate return eligibility using inventory and policy data
               Model : us.anthropic.claude-sonnet-4-5-20250929-v1:0

I'll help you with your return request for the wireless headphones. Let me first check your order details and eligibility.
Tool #1: get_inventory_context
Perfect! I can see your order details. Now let me verify your return eligibility:

Order Information:
- Order ID: ORD-27176
- Product: Wireless Headphones Pro
- Order Date: October 6, 2026
- Customer Tier: Premium (60-day return window)

Eligibility Check:
As a Premium tier customer, you have a 60-day return window from the order date. Your order was placed on October 6, 2026, which means you are well within the eligible return period.

Your return request is **APPROVED**. Let me initiate the return for you now.
Tool #2: initiate_refund
Return Successfully Initiated!

Your return has been approved and processed. Here are the details:

✅ **Return Reference Number:** RET-ORD-27176-1792327425
📦 **Order ID:** ORD-27176
🎧 **Product:** Wireless Headphones Pro
💰 **Refund Amount:** $149.99

Next Steps:
1. Please package the wireless headphones securely in their original packaging if possible
2. You will receive return shipping instructions via email at alice@example.com within 24 hours
3. Once we receive and inspect the item, your refund will be processed within 5-7 business days
4. Keep your return reference number (RET-ORD-27176-1792327425) for tracking purposes

Is there anything else you'd like to know about your return?   Status : [COMPLETE] Elapsed: 10.5s (LLM may still stream final text below)
               State : DynamoDB[refund_agent] version 1 -> 2
Perfect! Now let me draft the final customer response.
Tool #4: route_to_communication_agent

  [STEP 03] COMMUNICATION AGENT
               Role : Compose the final customer-facing response
               Model : us.anthropic.claude-sonnet-4-5-20250929-v1:0

Tool #1: get_full_workflow_context

Dear Alice,

Thank you for contacting us about your return request. I'm happy to help you with returning your Wireless Headphones Pro from order ORD-27176.

Good news! Your return has been successfully approved and initiated. ✅

Here's what you need to know:

Return Details:
- **Return Reference Number:** RET-ORD-27176-1792327425
- **Product:** Wireless Headphones Pro
- **Refund Amount:** $149.99

What happens next:

1. **Package your item** - Please securely pack the wireless headphones, preferably in their original packaging with all accessories included.

2. **Watch your email** - Within the next 24 hours, you'll receive detailed return shipping instructions at alice@example.com.

3. **Ship it back** - Follow the instructions in the email to send your item back to us.

4. **Get your refund** - Once we receive and inspect the headphones, we'll process your refund of $149.99 within 5-7 business days.

Important: Please keep your return reference number **RET-ORD-27176-1792327425** handy for tracking your return.

As a valued Premium member with us since 2021, we truly appreciate your business and want to ensure this return process is as smooth as possible for you.

Is there anything else I can help you with regarding your return or any other questions you may have?

Best regards,
Customer Support Team   Status : [COMPLETE] Elapsed: 8.9s (LLM may still stream final text below)
               State : DynamoDB[communication_agent] version 2 -> 3

Your return has been successfully processed! ✅

Return Reference Number: RET-ORD-27176-1792327425

Here's a summary of what happens next:

1. **Package your Wireless Headphones Pro** securely in its original packaging if possible
2. **Check your email** (alice@example.com) within 24 hours for return shipping instructions
3. **Ship the item back** using the provided label
4. **Receive your refund** of $149.99 within 5-7 business days after we receive and inspect the item

Please keep your return reference number for tracking purposes. As a valued Premium customer, we appreciate your business and want to make this process as smooth as possible for you.

Is there anything else I can help you with?Response: ---

Your return has been successfully processed! ✅
```

```
(venv) PS D:\Learning\future_aws_agent_engineer\multi_agent_customer_service_system> python src/agent_orchestrator.py test
**Your return has been successfully processed!** ✅

**Return Reference Number:** RET-ORD-27176-1792327425

Here's a summary of what happens next:

1. **Package your Wireless Headphones Pro** securely in its original packaging if possible
2. **Check your email** (alicc@example.com) within 24 hours for return shipping instructions
3. **Ship the item back** using the provided label
4. **Receive your refund** of $149.99 within 5-7 business days after we receive and inspect the item

Please keep your return reference number for tracking purposes. As a valued Premium customer, we appreciate your business and want to make this process as smooth as possible for you.

Is there anything else I can help you with?


  X-Ray trace 1-6a04bef0-049ca328d7f32aeb238b2918 published successfully.
  Allow 30-60 seconds, then open the Service Map and select 'last 5 minutes':
  https://console.aws.amazon.com/cloudwatch/home?region=us-east-1#xray:service-map/map
  Submission step: take a screenshot showing the full NovaMart-Orchestrator → worker-agent call chain.

──────────────────────────────────────────────────

Session: 9ebada01 | Customer: CUST-002
Query: What is the return policy for premium customers?
I'll help you with information about the return policy for premium customers. Let me initialize the session and retrieve that information for you.
Tool #5: initialize_session

Tool #6: route_to_policy_agent

  ──────────────────────────────────────────────────
  [ STEP 04 ]  POLICY AGENT
               Role  : Multi-Agent RAG -- parallel KB retrieval + synthesis
               Model : us.anthropic.claude-sonnet-4-5-20250929-v1:0
  ──────────────────────────────────────────────────

Tool #1: search_all_policies
    [PARALLEL RETRIEVAL - START]  Spawning 3 sub-agents concurrently via ThreadPoolExecutor
    ──────────────────────────────────────────────────
    +-- [SUB-AGENT 01]  ReturnsRetriever   KB-ID: F8BBEIKAUK
Tool #1: [SUB-AGENT 02]  ShippingRetriever  KB-ID: AXGEYKZN7C
    [PARALLEL RETRIEVAL - START]  Spawning 3 sub-agents concurrently via ThreadPoolExecutor
    ──────────────────────────────────────────────────
    +-- [SUB-AGENT 01]  ReturnsRetriever   KB-ID: F8BBEIKAUK
    +-- [SUB-AGENT 02]  ShippingRetriever  KB-ID: AXGEYKZN7C
    +-- [SUB-AGENT 03]  WarrantyRetriever  KB-ID: YPOXVVBRAH
    Retrieving in parallel... (sub-agent output suppressed - clean results printed sequentially after [END])

Tool #1: retrieve_shipping_policy
Tool #1: retrieve_returns_policy
## policy for premium customers**:

## to ** Premium Customer Return Policy

**Extended PremiumPremium customers**:

## Return Customer Premium Customer Window:** Return Policy

**Extended Return Policy Benefits
- Premium Return customers receive a ** Window:**

**Extended Return60-day return window** (
- Premiumcompared customers receive an Window:**
- Premium to  ** customers receive a30 days for Standard customers)extended **

##60-day return window** ( 60-day return window** ( Howcompared to Qualifycompared to the to  for Premium Status30 days for standard

You customers) standard 30-day window for automatically

**How regular become to Qualify a customers) for

## Premium customer when you Premium How meet either Tier:**
- Spend $500 to Qualify of these requirements+ in a calendar year, for Premium Status: OR
- Place 20+ orders

To become
- in a calendar year a Premium Spend **
- Automatic customer and$500  upgrade when accessor more** in a calendar year, threshold these OR
- Place **20  is met

**Additional benefits,or more orders** in a calendar year Premium you need

## Additional Benefits:**
- Free to meet Premium expedited shipping on all orders
- one Benefits

Beyond 3-year warranty on electronics ( of the following requirements the extendedvs.: return window, Premium customers also 1-year for standard)
- Spend ** receive:
- **Free expedited
- Priority customer support with$500  responseor more** in a calendar year, shipping** on all orders
- **3 within 4 hours
- Early OR
- Place **20  access to sales and new product launchesor more orders** in a calendar year-year warranty on electronics** (vs.
- Dedicated account manager for orders over

## Additional 1 $500

**Return Premium Benefits

Along-year for Standard customers)
- Process ( with the extended return period **Priority customer support** withSame, Premium customers also for receive:
- Free All response within 4 hours
- Early expedited shipping on all orders
- access to sales and new product launches Customers 3-year warranty on electronics (
-):**
1. Log in tovs. 1-year for standard Dedicated account manager for orders over $500)

## Important
- Priority customer support with Notes response within 4 hours
-
- Early access to sales and new product launches your account and go to Order History
2. Select the item and click "Start Return" Customers
3. Choose your return
- Dedicated account manager for orders are automatically upgraded when over $500

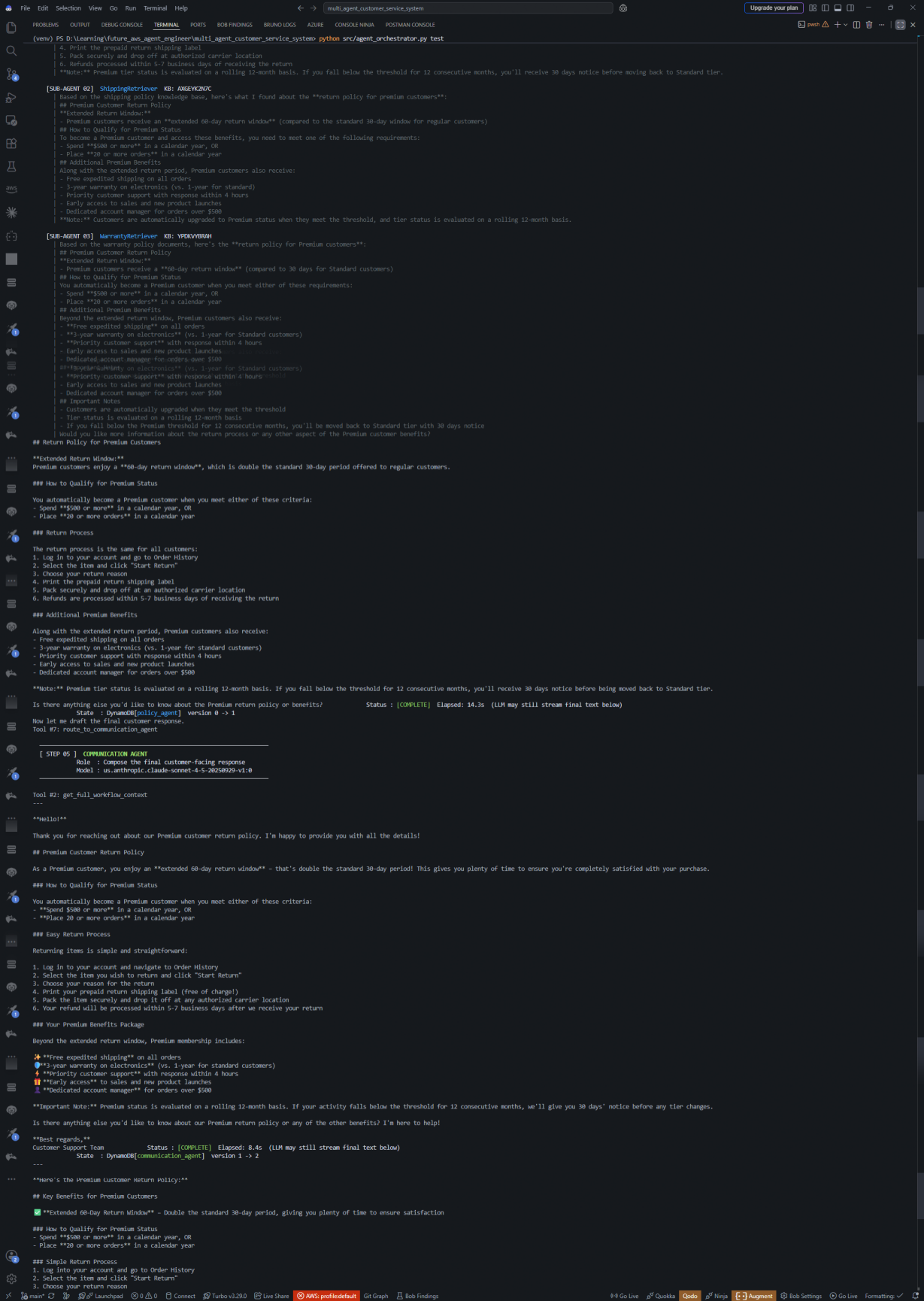
** reason they meet the threshold
4. Print the prepaid return shippingNote
- Tier status is evaluated on label
5. Pack securely and a rolling 12-month basis:** Customers drop off at authorized carrier location
- If you
6. Refunds processed within are fall below the Premium threshold for 12 5-7 business days of automatically upgraded to Premium status consecutive months, you'll be moved receiving when they meet the threshold back to Standard tier with 30  the, and tierdays notice

Would return

**Note status is evaluated on a rolling 12 you like more:** Premium tier information status-month basis. about the is evaluated on a rolling 12- returnmonth basis. process or any other aspect If you of the Premium customer benefits fall below the threshold for 1
2 consecutive months, you'll receive? 30 days notice before moving back to Standard tier.  ──────────────────────────────────────────────────
    [PARALLEL RETRIEVAL - END]  All 3 KBs responded  Elapsed: 8.0s

    [SUB-AGENT 01]  ReturnsRetriever   KB: F8BBEIKAUK
    | Based on the returns policy information, here's what applies to **Premium customers**:
    | ## Premium Customer Return Policy Benefits
    | **Extended Return Window:**
    | - Premium customers receive a **60-day return window** (compared to 30 days for standard customers)
    | **How to Qualify for Premium Tier:**
    | - Spend $500+ in a calendar year, OR
    | - Place 20+ orders in a calendar year
    | - Automatic upgrade when threshold is met
    | **Additional Premium Benefits:**
    | - Free expedited shipping on all orders
    | - 3-year warranty on electronics (vs. 1-year for standard)
    | - Priority customer support with response within 4 hours
    | - Early access to sales and new product launches
    | - Dedicated account manager for orders over $500
    | **Return Process (Same for All Customers):**
    | 1. Log in to your account and go to Order History
    | 2. Select the item and click "Start Return"
    | 3. Choose your return reason
```

Upgrade your plan



File Edit Selection View Go Run Terminal Help
multi_agent_customer_service_system
Upgrade your plan
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS BOB FINDINGS BRUNO LOGS AZURE CONSOLE NINJA POSTMAN CONSOLE
pwsh
(venv) PS D:\Learning\future_aws_agent_engineer\multi_agent_customer_service_system> python src/agent_orchestrator.py test
4. Print the prepaid return shipping label
5. Pack securely and drop off at authorized carrier location
6. Refunds processed within 5-7 business days of receiving the return
Note: Premium tier status is evaluated on a rolling 12-month basis. If you fall below the threshold for 12 consecutive months, you'll receive 30 days notice before moving back to Standard tier.
[SUB-AGENT 02] ShippingRetriever KB: AXGEYK2N7C
Based on the shipping policy knowledge base, here's what I found about the **return policy for premium customers**:
Premium Customer Return Policy
Extended Return Window:
- Premium customers receive an **extended 60-day return window** (compared to the standard 30-day window for regular customers)
How to Qualify for Premium Status
To become a Premium customer and access these benefits, you need to meet one of the following requirements:
- Spend **$500 or more** in a calendar year, OR
- Place **20 or more orders** in a calendar year
Additional Premium Benefits
Along with the extended return period, Premium customers also receive:
- Free expedited shipping on all orders
- 3-year warranty on electronics (vs. 1-year for standard)
- Priority customer support with response within 4 hours
- Early access to sales and new product launches
- Dedicated account manager for orders over $500
Note: Customers are automatically upgraded to Premium status when they meet the threshold, and tier status is evaluated on a rolling 12-month basis.
[SUB-AGENT 03] WarrantyRetriever KB: YPDKVVBRAH
Based on the warranty policy documents, here's the **return policy for Premium customers**:
Premium Customer Return Policy
Extended Return Window:
- Premium customers receive a **60-day return window** (compared to 30 days for Standard customers)
How to Qualify for Premium Status
You automatically become a Premium customer when you meet either of these requirements:
- Spend **$500 or more** in a calendar year, OR
- Place **20 or more orders** in a calendar year
Additional Premium Benefits
Beyond the extended return window, Premium customers also receive:
- **Free expedited shipping** on all orders
- **3-year warranty on electronics** (vs. 1-year for Standard customers)
- **Priority customer support** with response within 4 hours
- Early access to sales and new product launches
- Dedicated account manager for orders over $500
Important Notes
- Customers are automatically upgraded when they meet the threshold
- Tier status is evaluated on a rolling 12-month basis
- If you fall below the Premium threshold for 12 consecutive months, you'll be moved back to Standard tier with 30 days notice
Would you like more information about the return process or any other aspect of the Premium customer benefits?
Return Policy for Premium Customers
Extended Return Window:
Premium customers enjoy a **60-day return window**, which is double the standard 30-day period offered to regular customers.
How to Qualify for Premium Status
You automatically become a Premium customer when you meet either of these criteria:
- Spend **$500 or more** in a calendar year, OR
- Place **20 or more orders** in a calendar year
Return Process
The return process is the same for all customers:
1. Log in to your account and go to Order History
2. Select the item and click "Start Return"
3. Choose your return reason
4. Print the prepaid return shipping label
5. Pack securely and drop off at an authorized carrier location
6. Refunds are processed within 5-7 business days of receiving the return
Additional Premium Benefits
Along with the extended return period, Premium customers also receive:
- Free expedited shipping on all orders
- 3-year warranty on electronics (vs. 1-year for standard customers)
- Priority customer support with response within 4 hours
- Early access to sales and new product launches
- Dedicated account manager for orders over $500
Note: Premium tier status is evaluated on a rolling 12-month basis. If you fall below the threshold for 12 consecutive months, you'll receive 30 days notice before being moved back to Standard tier.
Is there anything else you'd like to know about the Premium return policy or benefits? Status : [COMPLETE] Elapsed: 14.3s (LLM may still stream final text below)
State : DynamoDB[policy_agent] version 0 -> 1
Now let me draft the final customer response.
Tool #7: route_to_communication_agent
[STEP 05] COMMUNICATION AGENT
Role : Compose the final customer-facing response
Model : us.anthropic.claude-sonnet-4-5-20250929-v1:0
Tool #2: get_full_workflow_context

Hello!
Thank you for reaching out about our Premium customer return policy. I'm happy to provide you with all the details!
Premium Customer Return Policy
As a Premium customer, you enjoy an **extended 60-day return window** - that's double the standard 30-day period! This gives you plenty of time to ensure you're completely satisfied with your purchase.
How to Qualify for Premium Status
You automatically become a Premium customer when you meet either of these criteria:
- **Spend $500 or more** in a calendar year, OR
- **Place 20 or more orders** in a calendar year
Easy Return Process
Returning items is simple and straightforward:
1. Log in to your account and navigate to Order History
2. Select the item you wish to return and click "Start Return"
3. Choose your reason for the return
4. Print your prepaid return shipping label (free of charge!)
5. Pack the item securely and drop it off at any authorized carrier location
6. Your refund will be processed within 5-7 business days after we receive your return
Your Premium Benefits Package
Beyond the extended return window, Premium membership includes:
✨ **Free expedited shipping** on all orders
🛡️ **3-year warranty on electronics** (vs. 1-year for standard customers)
📞 **Priority customer support** with response within 4 hours
🎁 **Early access** to sales and new product launches
👤 **Dedicated account manager** for orders over $500
Important Note: Premium status is evaluated on a rolling 12-month basis. If your activity falls below the threshold for 12 consecutive months, we'll give you 30 days' notice before any tier changes.
Is there anything else you'd like to know about our Premium return policy or any of the other benefits? I'm here to help!
Best regards,
Customer Support Team Status : [COMPLETE] Elapsed: 8.4s (LLM may still stream final text below)
State : DynamoDB[communication_agent] version 1 -> 2

Here's the Premium Customer Return Policy:
Key Benefits for Premium Customers
✅ **Extended 60-Day Return Window** - Double the standard 30-day period, giving you plenty of time to ensure satisfaction
How to Qualify for Premium Status
- Spend **$500 or more** in a calendar year, OR
- Place **20 or more orders** in a calendar year
Simple Return Process
1. Log into your account and go to Order History
2. Select the item and click "Start Return"
3. Choose your return reason
main* Launchpad 0 0 Connect Turbo v3.29.0 Live Share AWS: profile:default Git Graph Bob Findings Go Live Quokka Qodo Ninja Augment Bob Settings Go Live Formatting:

```
(venv) PS D:\Learning\future_aws_agent_engineer\multi_agent_customer_service_system> python src/agent_orchestrator.py test
4. Print your prepaid return shipping label
5. Pack securely and drop off at an authorized carrier
6. Refund processed within 5-7 business days

### Additional Premium Perks
- 🚚 **Free expedited shipping** on all orders
- 🛡️ **3-year warranty on electronics** (vs. 1-year standard)
- ⚡ **Priority support** – Response within 4 hours
- 🎁 **Early access** to sales and new products
- 👤 **Dedicated account manager** for orders over $500

**Note:** Premium status is evaluated on a rolling 12-month basis. You'll receive 30 days' notice if your tier changes.

Is there anything else you'd like to know about the Premium return policy or other benefits?Response: ---

**Here's the Premium Customer Return Policy:**

## Key Benefits for Premium Customers

✅ **Extended 60-Day Return Window** – Double the standard 30-day period, giving you plenty of time to ensure satisfaction

### How to Qualify for Premium Status
- Spend **$500 or more** in a calendar year, OR
- Place **20 or more orders** in a calendar year

### Simple Return Process
1. Log into your account and go to Order History
2. Select the item and click "Start Return"
3. Choose your return reason
4. Print your prepaid return shipping label
5. Pack securely and drop off at an authorized carrier
6. Refund processed within 5-7 business days

### Additional Premium Perks
- 🚚 **Free expedited shipping** on all orders
- 🛡️ **3-year warranty on electronics** (vs. 1-year standard)
- ⚡ **Priority support** – Response within 4 hours
- 🎁 **Early access** to sales and new products
- 👤 **Dedicated account manager** for orders over $500

**Note:** Premium status is evaluated on a rolling 12-month basis. You'll receive 30 days' notice if your tier changes.

Is there anything else you'd like to know about the Premium return policy or other benefits?


  X-Ray trace 1-6ad4bf12-12b7144f982592d7940770a1 published successfully.
  Allow 30-60 seconds, then open the Service Map and select 'Last 5 minutes':
  https://console.aws.amazon.com/cloudwatch/home?region=us-east-1#xray:service-map/map
  Submission step: take a screenshot showing the full NovaMart-Orchestrator → worker-agent call chain.

──────────────────────────────────────────────────────────────────────────────
Session: db14f790 | Customer: CUST-003
Query: How much would 5 items at $29.99 be with a 10% discount?
I'll help you with that calculation. Let me initialize the session first.
Tool #8: initialize_session
Based on your question, here's the calculation:

**Price Breakdown:**
- 5 items x $29.99 = **$149.95** (subtotal)
- 10% discount = **$15.00** (savings)
- **Final Total: $134.95**

This is a straightforward pricing calculation rather than a customer service request that requires routing to our specialized agents. If you have questions about applying a discount code to an actual order, need help with a purchase, or have other customer service needs, I'd be happy to assist you further!

Is there anything else I can help you with?Response: Based on your question, here's the calculation:

**Price Breakdown:**
- 5 items x $29.99 = **$149.95** (subtotal)
- 10% discount = **$15.00** (savings)
- **Final Total: $134.95**

This is a straightforward pricing calculation rather than a customer service request that requires routing to our specialized agents. If you have questions about applying a discount code to an actual order, need help with a purchase, or have other customer service needs, I'd be happy to assist you further!

Is there anything else I can help you with?


  X-Ray trace 1-6ad4bf2e-0a86b9bef1a584aa999c73e4 published successfully.
  Allow 30-60 seconds, then open the Service Map and select 'Last 5 minutes':
  https://console.aws.amazon.com/cloudwatch/home?region=us-east-1#xray:service-map/map
  Submission step: take a screenshot showing the full NovaMart-Orchestrator → worker-agent call chain.
(venv) PS D:\Learning\future_aws_agent_engineer\multi_agent_customer_service_system>
```